Why is product classification important for statutory accounting?

Source: SVILAC Ch. 1: Valuation Overview and Product Classifications

Product classification determines:

1. **Revenue and costs** reporting in income statement
2. **Methodologies and assumptions** for policy reserves
3. **Annual statement** exhibits, schedules, and supporting info

List four classifications under SSAP No. 50.

# Source: SVILAC Ch. 1: Valuation Overview and Product Classifications

1. **Life contracts** – financial assistance to beneficiary when insured dies
   - Long coverage period; mortality risk $\uparrow$ with age
   - Examples: WL, endowments, term, UL, variable life, annuities
2. **Deposit-type contracts** – no insurance risk
   - Financial/investment instruments (like banks)
   - Examples: annuities certain, lottery payouts, dividend accumulations
3. **A&H contracts** – protection vs. economic loss from medical condition
4. **P&C contracts** – protection vs. physical/financial damage to property or injury

What does SSAP No. 51 cover?

**Income recognition and reserving for life contracts**

- Recognize premiums as revenue on **gross basis**
  - **Gross premium** = amount charged to policyholder
  - **Net premium** = amount used in stat reserves
  - **Loading** = Gross – Net Premium
- Stat reserves required for all unmatured contracts
- Methods/assumptions: Appendices A-820 (SVL), A-822 (AOMR), C (AGs)
- Reserves must comply with ASOPs from ASB

Define RBC ratio and list 5 major risk categories.

# Source: SVILAC Ch. 1: Valuation Overview and Product Classifications

$$\text{RBC ratio} = \frac{\text{Total Adjusted Capital}}{\text{Life Insurer's RBC}}$$

**5 major risk categories:**

1. **C-0: Asset Risk – Affiliates** – default risk in affiliated investments
2. **C-1: Asset Risk – Other** – default risk (debt) — market losses (equity)
3. **C-2: Insurance Risk** – adverse mortality/morbidity
4. **C-3: Interest Rate, Health & Market Risk**
   - Reinvestment rates ¡ guaranteed rates
   - Capital losses (sale price ¡ cost)
   - Losses on variable product guarantees from market return changes
5. **C-4: General Business Risk** – fraud, mismanagement, other risks

What is the net level premium reserve method?

**Net premiums are constant % of gross premiums**

$$
\begin{aligned}
{}_tV_x^{NLP} &= \text{PVFB}_t - NP_0 \cdot \ddot{a}_{x+t} \\
NP_0 = PB_0 &= \left(\frac{\text{PVFB}_0}{\ddot{a}_x}\right) = \text{NP for first policy year} \\
r_t^{GP} &= \text{gross premium ratio} = \frac{GP_t}{GP_0} \\
\ddot{a}_x &= 1 + v \cdot {}_1p_x \cdot r_1^{GP} + v^2 \cdot {}_2p_x \cdot r_2^{GP} + \cdots \\
\ddot{a}_{x+t} &= r_t^{GP} + v \cdot {}_1p_{x+t} \cdot r_{t+1}^{GP} + v^2 \cdot {}_2p_{x+t} \cdot r_{t+2}^{GP} + \cdots \\
NP_t = PB_t &= PB_0 \cdot r_t^{GP}
\end{aligned}
$$

What is the full preliminary term (FPT) reserve method?

# Source: SVILAC Ch. 11: Valuation Methodologies

**FPT = Modified NLP with formulaic expense allowance**

$$
\begin{aligned}
{}_tV_x^{FPT} &= {}_tV_x^{NLP} - {}_tVE_x = \mathrm{PVFB}_t - (\mathrm{PVPB}_t + \mathrm{PVPE}_t) \\
{}_tVE_x = \mathrm{PVPE}_t &= PE_0 \times \ddot{a}_{x+t} \\
PE_0 &= \frac{EA_x}{\ddot{a}_x} \\
EA_x &= NP_1 - c_x = \left(\frac{\mathrm{PVFB}_1}{\ddot{a}_{x+1}}\right) - c_x \\
c_x &= v \cdot q_x \cdot \mathrm{DB} = \text{first-year COI} \\
NP_t &= \begin{cases} c_x & \text{for } t = 0 \quad (\alpha) \\ PB_t + PE_t = \dfrac{\mathrm{PVFB}_1}{\ddot{a}_{x+1}} \cdot r_t^{GP} & \text{for } t \geqslant 1 \quad (\beta) \end{cases}
\end{aligned}
$$

What is the commissioner’s reserve valuation method (CRVM)?

# Source: SVILAC Ch. 11: Valuation Methodologies

- Modified reserve where EA is **smaller** of:
  1. EA under FPT for contract
  2. EA under FPT assuming 20-pay contract
- Smallest reserves allowed by SVL
- ${}_tV_x^{\mathrm{CRVM}} = {}_tV_x^{\mathrm{FPT}}$ unless $m < 20$ years

Formulas for mean and mid-terminal reserves?

**Mean Reserve – DPA = Mid-Terminal Reserve + UPL**

$$
\begin{aligned}
\text{InterpMeanV} &= (1-h)\left({}_{t-1}V_x + NP_t\right) + h\left({}_tV_x\right) \\
\text{MeanV} &= \frac{\left({}_{t-1}V_x + NP_t\right) + \left({}_tV_x\right)}{2} \text{ if } h = 0.5 \text{ (common)} \\
\text{DPA} &= \sum \text{Modal NPs due between val date and next anniversary} \\
\text{InterpMidV} &= (1-h)\left({}_{t-1}V_x\right) + h\left({}_tV_x\right) \\
\text{MidV} &= \frac{{}_{t-1}V_x + {}_tV_x}{2} \text{ if } h = 0.5 \text{ (common)} \\
\text{UPL} &= \frac{\text{\# months until next premium}}{\text{\# months between premium payment}} \times \text{Modal NP}
\end{aligned}
$$

DPA and UPL held separately in stat financial statements

List types of continuous reserves.

# Source: SVILAC Ch. 11: Valuation Methodologies

1. **Semicontinuous**
    - DB paid at moment of death
    - Premiums paid BOY
2. **Fully Continuous**
    - DB paid at moment of death
    - NP paid continuously throughout year
    - UPL for NPs paid beyond val date
3. **Discounted Continuous**
    - Semicontinuous
    - Assumes refund of unearned premium at death
    - Requires DPA

What is the immediate payment of claims reserve (IPCR)?

Required for reserve methods assuming **curtate death benefits**

- Reality: DBs paid closer to time of death

$$\text{IPCR}_t = \begin{cases} \dfrac{i}{3}\text{PVFB}_{x+t} & \text{if DB paid \underline{without} interest} \\ \dfrac{i}{2}\text{PVFB}_{x+t} & \text{if DB paid \underline{with} interest} \end{cases}$$

Based on ”death portion of reserve”

List problems with CRVM addressed by principles-based approach.

# Source: ILA101-110: Fundamentals of the PBA to Stat Reserves

- **Rules-based and highly conservative** assumptions
  - Prescribed mortality and expense allowance
  - Interest rate based on formula
  - No explicit lapse assumption
- **Reserves don't vary with risk**
  - Same reserve for super preferred vs. standard class
- **CRVM unchanged for decades** despite product innovation
- Reg XXX and AXXX $\rightarrow$ excessive reserves for term and ULSG

What are the overarching principles of PBRs?

# Source: ILA101-110: Fundamentals of the PBA to Stat Reserves

**Overarching Principles**

1. Use **conservatism** reflecting:
   - Unfavorable events if reasonable probability
   - Tail risk (if significant)
2. **Consistency with company's risk management**
3. **Assumptions: prescribed and non-prescribed**
   - Non-prescribed = blend of company and industry experience
4. **Margins for uncertainty**
   - Greater uncertainty $\rightarrow$ larger margin $\rightarrow$ larger reserve

Also:

1. Ensure solvency
2. NOT be excessive
3. Reflect contracts' risk

List Valuation Manual objectives.

1. **Consolidate** min reserve requirements into one document
2. Promote **uniformity** among state valuation requirements
3. Provide **efficient, consistent & timely** process to update requirements
4. Mandate/facilitate **specific reporting** of experience data
5. Enhance **compliance** with SVL revisions

Outline the steps for determining VM-20 minimum reserve.

# Source: ILA101-110: Fundamentals of the PBA to Stat Reserves

1. **Calculate net premium reserve (NPR) floor** (ALWAYS for all policies)
    - Formulaic, prescribed assumptions (similar to CRVM)
2. **Calculate PBRs: deterministic and/or stochastic** (depends on exclusion tests)
    - Cash flow projection model
    - Assumptions = blend of company and industry experience
    - Determine in aggregate or by policy group, then allocate
    - Starting asset collar = 98% to max(NPR, 102%) of final reserve
    - Time period: until reserve no longer changes materially
3. **Excess Reserve = max[DR, SR] $-$ NPR**
    - For DR and SR, add $\pm$ allocated pre-tax IMR
    - Floored at 0
4. **Min Reserve = NPR + max[0, max(DR, SR) $-$ (NPR $-$ DPA)]**
5. **Aggregate min reserve = sum of min reserve for each product group**
    - Term, ULSG, other life

Steps for determining eligibility for exclusion tests?

## Source: ILA101-110: Fundamentals of the PBA to Stat Reserves

1. **Classify policies: term, ULSG, other life**
2. **Does group use clearly defined hedging strategy?**
   - If YES: Always do stochastic and deterministic (*done*)
   - Else $\rightarrow$ #3
3. **Decide if want stochastic exclusion test**
   - Any group: pass $\rightarrow$ no stochastic reserve required
4. **Other life only: decide if want deterministic exclusion test**
   - Only eligible if did stochastic exclusion test first
   - Pass $\rightarrow$ no deterministic reserve required
   - Term and ULSG must ALWAYS calculate deterministic reserves

What is the net premium reserve methodology?

$$\textbf{NPR} = \textbf{PV(Benefits)} - \textbf{PV(Net Premiums)}$$

**Floored by GREATER of:**

1. COI through next paid-to date
2. Cash value floor

**Differences vs. CRVM**

- Different EA
- Prescribed lapses for term and ULSG
- Prescribed mortality can be adjusted

**Similarities with CRVM**

- Prescribed CSO mortality
- Same interest rates
- Same gross premiums

What are the NPR’s objectives?

# Source: ILA101-110: Fundamentals of the PBA to Stat Reserves

- Floor gives **uniform baseline** for industry
- **Easier to audit** than DR or SR
- Only reserve required if pass **both exclusion tests**
- Can be used as **allocation basis** for DR/SR results

What is the deterministic PBR methodology?

**Determined in aggregate using <u>single</u> economic scenario**

1. **PV of Cash Flow Approach**

$$\underbrace{\text{APV(Ben, Exp \& Related)}}_{\text{Include existing PLs \& SA AV}} - \underbrace{\text{APV(Prems \& Related)}}_{\text{Include PL, reins, \& derivative CFs}}$$

   Discount rates vary by model segment (driven by assets in segment)

2. **Direct Iteration Approach**
   - DR = starting asset amount that matures all future benefits & expenses
   - Project all CFs iteratively until balance found

What is the stochastic PBR methodology?

**Determined in aggregate using same CFs as deterministic**

1. Project model segment CFs using stochastically generated scenarios
2. PV negative of projected statement value of assets
   - Discount rates = 1.05 $\times$ 1-year treasury scenario rates
3. Sum amounts in (2) across all model segments at end/beginning of each projection year
4. **Scenario reserve** = sum of starting assets across all segments + max of values above
5. **Stochastic reserve = CTE 70 of scenario reserve distribution**

General assumption considerations for deterministic and stochastic PBRs?

# Source: ILA101-110: Fundamentals of the PBA to Stat Reserves

- **Most assumptions = blend of company and industry experience**
  - Higher company data credibility $\rightarrow$ closer to company experience
  - "Prudent estimate"
- **Prescribed assumptions limited to:**
  - Interest rate movements (DR only)
  - Equity performance (DR only)
  - Spreads over treasuries on reinvestment
  - Definition of industry mortality table and grading method
  - ULSG: definition of industry lapse table
- **Margins for adverse deviation and estimate error (non-stochastic assumptions)**
  - Greater uncertainty $\rightarrow$ larger margin $\rightarrow$ larger reserve
  - Higher margin if credibility, data quality, assumption reliability are low

VM-20 mortality assumptions?

# Source: ILA101-110: Fundamentals of the PBA to Stat Reserves

- VM-20 prescribes company/industry blending table
  - Varies with credibility and ”sufficient data period”
- VM-20 prescribed margins vary by attained age
  - Company margins also vary with credibility
- Mortality improvement assumptions:
  - Between experience date and valuation date:
    - Improve company exp using supported company-specific factors
    - Improve industry exp using SOA-published rates
  - After valuation date (future mortality):
    - Use VM-20 prescribed improvement rates

VM-20 policyholder behavior assumptions?

**PH behavior risk factors:**

- Premium payment patterns and premium persistency
- Partial and full withdrawals (surrenders)
- Allocation between investment options
- Other PH benefit options

**VM-20 requirements:**

- Use experience from same block being modeled; else similar block
- Include margin: lower credibility $\rightarrow$ more conservative assumption
- Dynamic assumptions consistent with economic scenario

VM-20 expense assumptions?

# Source: ILA101-110: Fundamentals of the PBA to Stat Reserves

**Same assumption for DR and SR** (except inflation effects)

**Considerations:**

- Spread capital expenditures (IT, etc.)
- Assume going concern
- Expense basis
  - Align with actual expense
  - Consistent between new and in-force
- Reflect inflation
- NO future expense improvements
- NO taxes
- Consistency with related assumptions

**Expense types to include:**

1. Direct + indirect + overhead
2. Acquisition expenses
3. Expected future non-recurring
4. Efficiencies from combinations/mergers

**Use FULLY allocated expenses**

Compare reinsurance: rules-based vs. principles-based.

## Source: ILA101-110: Fundamentals of the PBA to Stat Reserves

**Rules-Based**

- Mirror-image reserves (usually)
- Ceding company reserve credit = reinsurer's assumed reserve

**Principles-Based**

- Mirror-image reserves not required (or likely)
- Ceding company and reinsurer use own assumptions
- Reins Reserve Credit = Pre-Reins Min Reserve − Post-Reins Min Reserve
  - Pre-reins run: follow VM-20 assuming reins didn't exist
  - 98% to 102% collar for starting assets does <u>not</u> apply
- Reins features like stop-loss may require additional stochastic analysis
- If YRT rates non-guaranteed: reserve credit based on pre-PBR approach

Overview of asset modeling framework under VM-20?

**Existing Assets = SA Assets + PLs + Derivatives + GA Assets**

- ”Solve” for GA Asset balance so total = 98% to 102% of reported reserve
- Requires iterations

**Reinvestment Assets**

- VM-20 prescribes yield spread
- Reflect investment policies of company
  - Should not produce lower reserve than ”alternative investment strategy”
  - VM-20 prescribes 50/50 mix of A and AA public non-callable corporate bonds

**Reflect default costs on all modeled assets**

List the criteria for a Clearly Defined Hedging Strategy (CDHS).

# Source: ILA101-110: Fundamentals of the PBA to Stat Reserves

1. Risks being hedged: cash flow, fee income, interest credits, etc.
2. Hedging objectives
3. Material risks <u>not</u> hedged: mortality, withdrawal, etc.
4. Financial instruments used
5. Trading rules and tolerances
6. Metrics and frequency for effectiveness measurement
7. Conditions and duration for non-hedging
8. Responsible group or area
9. Identified basis, gap, or assumption risks
10. Ineffectiveness circumstances

Overview of economic assumptions under VM-20?

# Source: ILA101-110: Fundamentals of the PBA to Stat Reserves

- Based on NAIC's Generator of Economic Scenarios (GOES)
- GOES generates Treasury rates and S&P 500 total returns
  - Based on prescribed parameters (same for all companies)
- GOES can generate up to 10,000 scenarios
  - Companies may use subset for SR
  - Subset use = simplification technique (must not materially understate reserves)
- DR scenario is No. 12 from stochastic exclusion test set

Outline the key roles and responsibilities for corporate governance under VM-20?

Source: ILA101-110: Fundamentals of the PBA to Stat Reserves

1. **Board of directors**
   - Oversee management's process to correct material control weaknesses
   - Oversee PBR infrastructure / consistency with company's risk management
   - Interacts with senior management
   - Document reviews and actions
2. **Senior management**
   - Establish PBR infrastructure and modeling resources
   - Review PBR assumptions, methods, and models
   - Review significant/unusual issues with PBR results
   - Adopt internal VM compliance controls
3. **Qualified actuary(ies)**
   - Must meet Academy's qualification standards
   - Oversees PBR calculation
   - Develops internal standards for actuarial practices, controls, documentation
   - Summary report to board and senior management
   - Disclose significant unresolved issues to auditors/regulators
   - Issue VM certifications but not necessarily opinion on reserve adequacy

Value of future experience submissions?

# Source: ILA101-110: Fundamentals of the PBA to Stat Reserves

- Helps regulators and reviewers
  - Easier to monitor, perform reasonableness checks, develop thresholds
- Facilitates credibility-based blends
- Provides database of inter-company experience
- Data source for standard valuation table updates

Two PBR support components implemented by NAIC?

# Source: ILA101-110: Fundamentals of the PBA to Stat Reserves

1. **Additional state actuarial resources**
   - Aid in reviewing PBRs and supporting updates requirements/examination tools
   - Facilitate refinement and implementation of PBR requirements
2. **Valuation Analysis Working Group (VAWG)**
   - Responds to states confidentially during annual PBR reviews/exams
   - Utilizes state actuarial resources to:
     - Assist with issues and questions
     - Maintain documentation/databases to increase support consistency
   - Provides analysis and recommendations for PBR issues to address in VM

GAAP definition, primary audience, and information sources?

# Source: GAAP Ch. 1: GAAP Objectives and Their Implications

**GAAP** = conventions, rules, and procedures defining accepted accounting practices

- Provides transparency to capital providers and related audiences
- Capital providers = **stockholders** and **policyholders**
  - Stock companies owned by stockholders
  - Mutual companies owned by policyholders
- **Financial statements** = primary GAAP information source
  - Income statement, balance sheet, etc.
- Other sources:
  - Conference calls with analysts
  - Responses to rule-makers' queries
  - Supplementary investor information
  - Press releases

Purpose of GAAP financial statements?

# Source: GAAP Ch. 1: GAAP Objectives and Their Implications

- Summarizes enterprise's assets, liabilities, and transaction effects on earnings
- Expressed in single, uniform currency unadjusted for inflation
  - Non-monetary info converted to monetary
- Measures **single entity**, not entire industry
  - Subsidiaries consolidated into one set of statements
- Reportable **segments** based on operating segments
  - Operating segment = relatively homogenous segment per company's decision-makers
  - May differ company to company
- Measured over clearly defined period (quarterly or annually)

List users of financial statements.

# Source: GAAP Ch. 1: GAAP Objectives and Their Implications

- **Capital providers (debt and equity)**
  - Financial institutions
  - Investment banks
  - Private lenders
  - Commercial banks
  - Individual investors
- **Brokers and dealers** in financial instruments issued by enterprise
- **Expert advisors** to users
  - Attorneys, actuaries, accountants
- **Policyholders** or other stakeholders

List parties with GAAP accounting oversight roles.

# Source: GAAP Ch. 1: GAAP Objectives and Their Implications

- **Management responsibilities:**
  - Fair presentation per GAAP
  - Complete financial reporting process
  - Explain matters not evident in numerical info
- **Auditors:** opine on fairness of presentation and internal control effectiveness
  - Independent external auditors – not employees (hired by/report to Board)
    - Opinions based on testing and COSO criteria
  - Internal auditors – can be employees or contracted third parties
- Other regulatory bodies: SEC, IRS, state insurance departments
- Users acting on behalf of others: stock analysts, rating agencies

Standard-setters for GAAP accounting?

# Source: GAAP Ch. 1: GAAP Objectives and Their Implications

- **Financial Accounting Standards Board (FASB)**
  - Establishes uniform standards for financial statement contents
  - Maintains Accounting Standards Codification (ASC)
  - Issues Statements of Financial Accounting Concepts (SFACs)
  - Standardization reduces need for individual specific info requests
- **AICPA** and **PCAOB:** issue standards for accountants and auditors
  - AICPA: American Institute of Certified Public Accountants
  - PCAOB: Public Company Accounting Oversight Board
- **SEC:** ultimate regulator of US public companies
  - Can de-list or halt stock trading
  - Issues pronouncements: manuals, staff bulletins, enforcement releases

Meaning and importance of relevance and reliability for GAAP?

# Source: GAAP Ch. 1: GAAP Objectives and Their Implications

**Financial statement info must be relevant and reliable**

- **Relevant** info must be **timely**
  - Perfect info not valuable if too late
- **Relevant** = **consistent and comparable**
  - Consistent = based on similar/identical measures from prior periods
  - Comparable = equivalent to measures used in other enterprises
  - Should be comparable across time periods
  - Prior results restated when errors surface
  - Some standards entity-specific (e.g. ASC 944 for insurers only)
- **Relevant** = provides appropriate detail
  - Adequate detail informs without confusing
- **Reliable** = materially accurate and verifiable
- **Reliability** also implies ”free of bias”

Key relationships between GAAP balance sheet and comprehensive income?

# Source: GAAP Ch. 1: GAAP Objectives and Their Implications

**Balance Sheet: Assets = Liabilities + Equity**

- Equity = Assets − Liabilities
- Equity = Paid-In Stock + Paid-In Additional Capital + Retained Earnings
- EOY Equity = BOY Equity + Comprehensive Income + Owner Transactions
  - Owner Transactions = Capital Contributions − Dividends

**Comprehensive Income** = Net Income + Other Comprehensive Income (OCI)

- Income Statement: Net Income = Revenues − Expenses
  - Revenue = Premiums + Fees + InvInc + Realized Gains
  - Expenses = Benefits + Expenses − Realized Losses
- OCI = equity changes from gains/losses not central to operations
  - Changes in unrealized gains/losses on AFS assets
  - Shadow adjustments
  - MRB FV changes due to own credit risk

Define asset, liability, and equity per SFAC No. 8 (CON8 Ch4).

# Source: GAAP Ch. 1: GAAP Objectives and Their Implications

**SFAC No. 8 (”CON8 Ch4”) defines balance sheet elements:**

- **Asset** = present right of entity to economic benefit
  - Restricts others’ access to asset
  - Provisions for losses required if probable future benefit ↓
  - Includes tangible (e.g. financial instruments) and intangible (e.g. goodwill)
- **Liability** = present obligation of entity to transfer economic benefit
  - ASC 450 requires liabilities for contingent events that are probable and reasonably estimable
- **Equity** = residual interest in entity’s assets after deducting liabilities
  - Aka ”net assets”

SFAC No. 5 criteria for recognition and measurement?

# Source: GAAP Ch. 1: GAAP Objectives and Their Implications

1. **Definition:** asset, liability, revenue, or expense
2. **Measurability:** must have quantifiable attribute with sufficient reliability
   - Historical cost: easy via company records; adjust for depreciation
   - Fair value: from active secondary markets or expert appraisals
   - Present value: discount CFs at rate that can vary
     - Least preferable, but necessary for most ILA products

   Always use uniform monetary units <u>un</u>adjusted for inflation
3. **Relevance:** consistent, comparable, meaningful in users' decision-making
4. **Reliability:** accurate, verifiable, free of bias

Compare cash basis vs. accrual accounting.

# Source: GAAP Ch. 1: GAAP Objectives and Their Implications

- **Cash** basis is simplest
  - Record entries only when cash changes hands
  - Not appropriate for incomplete or long-term transactions (like life insurance)
- **Accrual** alters timing of when cash transactions are recognized
  - Goal: earnings emerge proportionally to ”completeness” of earnings process
  - Examples: reserves, DAC, unearned premium liabilities

What is the double-entry accounting system?

# Source: GAAP Ch. 1: GAAP Objectives and Their Implications

**GAAP predicated on double-entry accounting**

| Financial Statement | Account Type | Debit | Credit |
|---|---|---|---|
| Income Statement | Revenue | − | + |
| | Expense | + | − |
| Balance Sheet | Asset | + | − |
| | Liability or Equity | − | + |

**This system ensures A = L + E always holds**

- Debits = ↑ assets and expenses
- Credits = ↑ liabilities, equity, and revenue
- Credits and debits always balance in any period

Considerations for materiality in GAAP financial statements?

# Source: GAAP Ch. 1: GAAP Objectives and Their Implications

**Key tradeoff:** approximately correct, timely info $>$ more accurate info too late

**Estimates should be precise enough for informed decisions**

- Some users require more precision than others
- Material errors should be corrected
- Immaterial imprecision doesn't need correction
- "Immaterial" = impact won't influence decision-making
  - Requires judgment

List key disclosures required under GAAP.

## Source: GAAP Ch. 1: GAAP Objectives and Their Implications

- **Footnotes and narrative disclosures** supplement primary financial statements
  - E.g. describe future liability not currently quantifiable
- **SEC Regulation S-K** – market risk exposure impact on financial statements
  - Quantitative disclosures: market risk, earnings sensitivity
  - Qualitative info on exposures
- **Disclosures for accounting principle changes**
  - ASC 250: accounting changes and error corrections
  - ASC 855: significant events between period end and statement release
- **Additional GAAP disclosures:**
  - Company and product description
  - Summary of significant accounting policies
  - Details of invested assets
  - Details of reserves and accounting assets

List acquisition costs capitalized in DAC (traditional & LP).

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

**Deferrable = incremental direct costs of successful acquisition:**

- Employee comp tied to selling, UW, issue, processing issued contracts
- Other costs not incurred if the contract were not acquired
- Commissions $>$ ultimate (renewal) level
- LP: all premium-based commissions (ultimate level = 0)
- Certain direct marketing costs

Expense all else as incurred (overhead, general advertising, unsuccessful efforts)

**Ignore premium taxes & FIT in the LFPB and DAC** (exception: premium taxes on LP contracts may be deferrable)

Formulas for grouped DAC amortization and rollforward?

## Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

$$
\begin{aligned}
\text{AmortRatio}_t &= \frac{\text{BOY DAC}_t + \text{Cap}_t}{\sum_{s \geqslant t} \text{expected BOYIF}_s} \\
\text{Amort}_t &= \text{AmortRatio}_t \times \text{BOYIF}_t \\
\text{EOY DAC}_t &= \text{BOY DAC}_t + \text{Cap}_t - \text{Amort}_t - \text{True-up}_t
\end{aligned}
$$

- Yr 1: $\text{Amort}_1$ = Deferrable $\times$ $\text{BOYIF}_1$ / $\sum$BOYIF
- Constant-level basis: amount/units inforce (initial DB if non-level)
- <u>No interest accretion</u> on DAC

Describe the DAC experience true-up (unlocking).

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

$$\text{True-up}_t = \text{DAC}_t \times \frac{\text{expected BOYIF}_t - \text{actual BOYIF}_t}{\text{expected BOYIF}_t}$$

- Actual IF $<$ expected $\rightarrow$ write DAC down (extra amortization)
- One-sided: no adjustment if more persist than expected
- Revise future amort prospectively over the new inforce projection
- True-up is just extra amort; total amort + true-ups = deferrable

Considerations for cohort grouping and unit of account?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

**Contract grouping necessary to calculate NPR**

- NPR based on group experience/assumptions
- Reserves can still be calculated at contract level

**Groups must consist of same cohort** (issued within 1 year)

**Contracts in group should be similar by:**

- Type (e.g. don't mix term and WL)
- Contract length
- Premium paying period (don't mix LP and non-LP)
- Rating class
- Issue age (possibly banded)
- Face amount band

**More (smaller) groups $\rightarrow \uparrow$ assumption precision, BUT:**

- Credibility $\downarrow$
- NPR more likely to be capped

LFPB for traditional nonpar insurance?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

$$\begin{aligned}
\text{NPR}_t &= \frac{\text{PV}_0(\text{Actual Benefits Through } t) + \text{PV}_0(\text{Expected Benefits After } t)}{\text{PV}_0(\text{Actual GPs Through } t) + \text{PV}_0(\text{Expected GPs After } t)} \\
\text{LFPB}_t &= \text{PVFB}_t - \text{NPR}_t \times \text{PVGP}_t \\
&= (\text{LFPB}_{t-1} + \text{NPR}_t \times \text{GP}_t)(1+i) - \text{Ben}_t \\
&= \text{LFPB}_{t-1} + \text{NP}_t + \text{Accretion}_t - \text{Ben}_t
\end{aligned}$$

- NPR calculated at group level and capped at 100%
- ”Benefits” = DBs, maturities, CSV, and claim/termination-related expenses (LAE)
- CF assumptions = best estimates without PADs revised at least annually
- Discount rate = upper-medium grade yields (A-rated)
  - Maximize relevant observable inputs; minimize unobservable
  - Must reflect duration characteristics of liabilities
  - Can be curve or blended single rate
  - Measure LFPB at original (accretion rate) and current DR
- LFPB can’t be negative at group level: floor at 0

Rules for truing up NPR for traditional long-duration insurance?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

- **CF assumptions <u>not</u> locked in at issue**
- **Recalculate ("true-up") NPR from issue** (at least annually)

$$\text{NPR}_t = \frac{\text{PV}_0(\text{Actual Benefits Through } t) + \text{PV}_0(\text{Expected Benefits After } t)}{\text{PV}_0(\text{Actual GPs Through } t) + \text{PV}_0(\text{Expected GPs After } t)}$$

  - Replace estimated CFs with actuals
  - Must revise/re-confirm assumptions when CFs updated
  - Use original discount rate ("accretion interest rate")
- **NPR can be updated more frequently than annually** (e.g. quarterly)
  - Review at same time each year
  - Interim reviews may be needed:
    - Special situations: pandemics, medical advances, competitive changes
    - Other significant developments
  - Must update all CF components
  - Must address all assumptions (exception: LAE can be locked-in)

How to quantify and present remeasurement gains/losses?

ASC 944 requires separate **remeasurement loss/gain** line on income statement

Total effect on income when benefits deviate from expected:

$$\begin{aligned}
&\text{Actual Benefits Incurred} \\
&\quad + \text{Remeasurement Loss} \qquad (= \text{Revised BOP LFPB} - \text{Original BOP LFPB}) \\
&\quad + \text{Change in Reserve} \qquad (= \text{EOP LFPB} - \text{Revised BOP LFPB}) \\
&= \text{Total Benefit Expense}
\end{aligned}$$

Calculate all items above using at-inception DR (accretion interest rate)

How to present GAAP net income on income statement and in SOE format for traditional nonpar?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

**GAAP net income =**

**GAAP Revenue** = Gross Premiums + Net Investment Income

− **GAAP Expenses** = DBs + SBs + IncrLFPB + Non-Def Exp + DAC Amort

**SOE Format:**

| | **Earnings Source** | **Formula** |
|---|---|---|
| (+) | Premium Load | GP − NP |
| (+) | Interest Spread | NII − Accretion |
| (−) | Acquisition Costs | DAC Amort + Non-Def AcqCost |
| (−) | All Other Expenses | MaintExp + RenewalComm + PremTax + Overhead |
| (−) | Benefit Deviation | Actual Benefits − Expected Benefits |
| (+) | Other Reserve Unlock | IncrLFPB Before NPR Unlock − IncrLFPB After NPR Unlock<br>+ Int Spread Before NPR Unlock − Int Spread After NPR Unlock<br>+ Prem Load Before NPR Unlock − Prem Load After NPR Unlock |
| = | **GAAP Net Income** | |

DAC is <u>non-interest-bearing</u>: large early DAC $\downarrow$ invested assets $\rightarrow$ $\downarrow$ early interest spread $\rightarrow$ contributes to yr-1 loss (amort largest in yr 1)

ASOP 10 guidance for setting best estimate assumptions?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

- Use best estimates without conservatism or PADs
- Best estimate = ”most likely, average, or central” outcome
  - Corresponds to mode, mean, or median of probability distribution
  - Other interpretations possible
- Considerations when selecting best estimates:
  - Characteristics/magnitude of company’s business
  - Company maturity and growth rate
  - Company’s prior experience and trends
  - Medical, economic, social, technological developments
  - Actual recent company experience (if relevant and credible)
  - Relevant industry data to supplement company experience
  - ASOP 23 for data quality issues

Considerations for mortality and lapse assumptions for traditional nonpar?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

**Mortality**

- Typically based on company experience studies
- New products may require industry experience, adjusted as needed
- Reflect marketing and UW practices
- Reflect historical mortality improvement if actuary believes it will continue
- Can be dependent on lapses (anti-selection)

**Lapse rates**

- Account for face, issue year/age, duration, premium paying period/frequency
- Composite rates should account for current and future business mix

Assumption considerations for non-fixed premiums?

## Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

- **Most nonpar trad contracts have fixed premiums**
  - Fixed premiums = contract parameter, not assumption
- **Examples with adjustable premiums:**
  - Many term policies: premiums adjustable up to contract maximum
  - Indeterminate premium contracts: premiums reset prospectively if assumptions change
- Adjustable premiums require **best estimate assumption** considering:
  - Current premium levels
  - Historical premium patterns
  - Expectations of future increases/decreases
  - Contractually guaranteed maximum premiums
- **Include actual premiums when unlocking NPR**
- Indeterminate premium policies are **UL-type** if satisfying 1 of:
  - Stated account balance with interest credits/charges not fixed/guaranteed
  - Interest rates and market conditions mainly drive changes in contract elements

Treatment of coupons, dividends, and conversion privileges for traditional nonpar?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

- **LFPB should reflect coupons, endowments, and conversion privileges**
- **Dividend-paying policies not following contribution principle:**
  - Dividend revisions may follow different schedule than other assumption revisions
  - Dividend scale slope impacts LFPB
  - May create anomalies when dividend elements don't align with current experience
    - E.g. dividend investment return assumption may not match current returns
  - PH dividend option utilization usually not reflected in LFPB
    - Assume dividends always paid in cash
- **If shareholder dividends restricted, establish UPPEA**
  - Do not include PH dividends in LFPB when using UPPEA

Accounting for riders on traditional nonpar?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

**If rider can't exist standalone** (e.g. premium waiver, LTCI/CI combo):

- Treat base + rider as single unit
- Include rider charges and benefits in LFPB
- Include deferrable acquisition costs in DAC
- Riders often profitable $\rightarrow$ will lower NPR, all else equal

**Separate riders insuring risk other than base contract**

- Examples: disability income riders, spouse term riders
- Assumptions should be internally consistent with base
- In some situations, insurers set rider GAAP reserve = stat reserve
  - Appropriate if not materially different

GAAP valuation approaches for disability waiver riders?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

**If disability <u>not</u> occurred:**

1. **Gross approach** – value rider separately from base
   - Reserve = PV expected waived premiums $-$ NPR $\times$ PV expected rider charges
2. **Net approach** – integrate rider with base
   - Reduce base contract premiums for expected waived amounts

**Mortality assumption approaches:**

- Could use higher mortality for contracts with active disability
- Could use averaged mortality assumption for all contracts (disabled or not)
- Consider credibility/availability of disabled life mortality data

Rules for updating discount rates on traditional nonpar?

## Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

**Update <u>every</u> reporting period (even if not addressing CF assumptions)**

**Change in LFPB from new discount rates $\rightarrow$ OCI**

$$\begin{aligned} \text{OCI}_t &= \text{AOCI}_t - \text{AOCI}_{t-1} \\ \text{AOCI}_t &= \text{accumulated OCI at time } t \\ &= \text{LFPB}_t \text{ based on original DR} - \text{LFPB}_t \text{ based on current DR} \end{aligned}$$

**Do NOT recalculate NPR using current DR**

- Always use NPR from most recent CF experience/assumption unlocking

**OCI accounting reduces GAAP equity volatility**

- LFPB on balance sheet based on current discount rates
- When interest rates $\uparrow$: AFS asset FV and LFPB "FV" $\downarrow$
- Reductions in AFS asset FV $\rightarrow$ negative OCI
- Reductions in LFPB "FV" $\rightarrow$ positive OCI

Compare 3 approaches for locked-in interest accretion when using discount rate curve.

Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

**3 approaches for locking in curve of accretion rates:**

1. **Spot rate** – use original spot rates for each CF through time
2. **Forward rate** – use implied period-specific forward rates
3. **Effective yield** – use IRR based on spot curve

Differences:

- Effective yield may exceed long-term spot rates if premiums payable over multiple years and yield curve upward sloping
  - From ”leveraging effect”: premiums ¿ benefits in early years, then reverses
  - Can cause spot rate accretion ¿ effective yield method
- Spot rate interest accretion may better match investment income
- Forward rate usually produces least accretion in early years
  - Accretion in later years very high (may exceed investment income)
  - Also $\rightarrow$ lower early reserves
  - Forward curves tend to dip/spike $\rightarrow$ affects interest accretion

Practical challenges and solutions for multiple issue dates within cohort?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

**Locked-in DRs must reflect rate environment when contracts sold**

- Challenge: interest rates vary throughout issue year
- May prevent using single yield curve at BOY/MOY/EOY

**Possible solutions**

- Use average curve
  - Could be weighted (e.g. by premium) or simple
  - Could average only key dates (e.g. month-ends)
- Use provisional yield curve(s)
  - Lock in final yield curve(s) at EOY
- Set DRs by sub-cohorts (e.g. monthly, quarterly)
  - Avoids provisional yield curves
  - NPR and cap still set at <u>cohort</u> level

How to apply ASC 820 guidance for unobservable discount rates?

**Follow ASC 820 level 3 measurement:**

- Reflect market participants' assumptions to price assets/liabilities, including risk
- Reflect risk in both valuation technique and inputs
- Reduce DRs by margin to reflect significant uncertainty from lack of market activity
- Insurer's own data should be adjusted to reflect market participant's perspective
  - Doesn't need to be exhaustive effort

Approaches when no A-rated rates for foreign currency:

- Use highest available rating instead
- Subtract assumed spread from highest rated rates

Techniques for estimating long-dated interest rates?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

1. Interpolate between last observable rate and assumed LT rate
2. Determine forward rate from last observable rate and extrapolate
3. Add A-rated spread for shorter rates to longer risk-free rates
   - Often risk-free rate curve extends beyond A-rated curve
   - Assume A-rated spreads over risk-free rates constant
   - Can augment either other method

Nature of limited-pay contracts and how they're accounted for?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

**LP contracts have premium payment periods ¡ coverage period**

- Single-pay, 10-pay, 20-pay, payments to age 65, etc.
- Also includes life payout annuities (usually single premium)
- Problem: high profit during premium period, then losses

**Solution: defer GP in excess of NP**

- Life insurance: DPL amortized vs. discounted inforce
  - If experience = assumptions, profits emerge as level % of inforce
- Payout annuities: DPL amortized vs. PV future benefits
- **Use same CF assumptions and accretion rates as LFPB**
  - Update experience/assumptions when LFPB updated
  - Do not update discount rate

How to determine DPL for limited-pay traditional nonpar?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

1. Determine annual profit to defer

$$\text{Profit}_t = (1 - \text{NPR}) \times \text{GP}_t$$

2. Calculate DPL as constant % of inforce

$$\begin{aligned}\text{DPL\%} &= \frac{\text{PV}_0(\text{Profit})}{\text{PV}_0(\text{BOYIF}^*)} \\ \text{DPL}_t &= \text{DPL\%} \times \text{PV}_t(\text{BOYIF}^*) - \text{PV}_t(\text{Profit}) \\ &= (\text{DPL}_{t-1} + \text{Profit}_t) \times (1 + i) - \text{DPL\%} \times \text{BOYIF}^*_{t-1}\end{aligned}$$

Discount BOYIF* as of EOY

3. Include ↑ in DPL on income statement as expense
   - As profit capitalized, DPL ↑ to offset and defer it
   - DPL ↓ each period after premiums end releasing profit into income

Define restricted dividends.

**Shareholder dividends may be restricted by law/charter/policy form**

- Protects PHs' share of profits (future PH dividends)
- PH surplus share excluded from stockholder equity

**When restricted dividends exist, insurer must establish UPPEA**

- UPPEA = deferred surplus (profits) paid as future PH dividends
- Smooths net income $\rightarrow$ more even stream of shareholder dividends
- Do not include dividends in LFPB

How to calculate UPPEA?

## Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

Assuming shareholder dividends restricted to 10% of surplus:

1. Project GAAP income statement
2. Calculate GAAP net income ignoring PH dividends
3. Calculate running total of GAAP surplus before PH dividends

$$\text{CumSurplus}_t = \text{CumSurplus}_{t-1} + \text{NetIncomeBefore}_t$$

4. Calculate running total of PH dividends paid
5. $\text{UPPEA}_t = 0.90 \times \text{CumSurplus}_t - \text{CumPHDivs}_t$
6. Calculate final GAAP net income:

$$\text{NetIncome}_t = \text{NetIncomeBefore}_t - \text{PHDivs}_t - \Delta\text{UPPEA}_t$$

Shadow UPPEA may be needed to offset OCI from AFS assets

- Changes in shadow UPPEA $\rightarrow$ OCI

Examples and characteristics of claims liabilities?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

Claims liabilities established for:

- Pending death claims
- Resisted claims
- Incurred but not reported (IBNR)

**Characteristics:**

- Usually not discounted (paid out soon)
- Less significant for life insurance (claims paid quickly)
- IBNR can be shown separately on balance sheet or included with LFPB
- Allocate change in total IBNR to cohorts using ”rational allocation” method
  - Liabilities for pending/resisted claims tied to original cohort more easily

Accounting for premium accruals?

**Reported separately on GAAP balance sheet:**

- Advance premium liability – offsets premiums paid in advance
- Receivable asset for due/unpaid premiums – $\uparrow$ income in place of premiums

**Netted against or added GAAP reserve (don't report separately):**

- DPA for non-annual modes (relevant to mean reserves)
- UPL (relevant to midterminal reserves)

Challenges in explaining change in LFPB to others?

- New policies more sensitive (more affected by PV effects)
  - Opposite for older policies ($\downarrow$ future CFs remaining)
- Reserve sensitivity $\uparrow$ with NPR (max sensitivity at 100% cap)
- Many experience/assumption changes impact both benefits and premiums
- Deviations in current experience also impact future experience
- If experience only updated annually, impact affected by deviations in prior periods
- $1 - \text{NPR}$ = insurer's best estimate profit margin (excluding expenses)
  - Changes every time NPR changes

Considerations for interim reporting?

**Interim reporting** – many insurers report GAAP earnings quarterly

- Will $\uparrow$ volatility for insurers not updating NPR each quarter
- Prior quarterly results can't be restated at EOY (must stand on their own)
- Will likely cause results to differ from annual-only reporting

Disaggregated rollforward disclosure requirement for traditional nonpar?

# Source: GAAP Ch. 5: Nonparticipating Traditional Life Insurance

**Key disclosure:** YTD disaggregated tabular rollforward of GAAP reserve from beginning to ending balance

- Must be gross of reinsurance recoverables and include any DPL
- Show future NPs and expected benefits separately
- Aggregation level should align with insurer's business
  - Separate balances by geography, LOB, distribution channel, etc.

**Examples of potentially relevant items:**

- Newly issued contracts
- Accrued interest and NPs collected
- Benefit and expense payments
- Derecognition of contracts due to death or lapse
- Details around true-ups, unlocking, discount rate updates, etc.

IFRS 17 scope, general measurement model, and effective date?

# Source: ILA101-111: Insurance Contracts First Impressions

**Scope:** contracts meeting IFRS 17 definition of insurance contract

**General measurement model**

- Liability = Fulfillment Cash Flows (FCF) + Contractual Service Margin (CSM)
- FCF = PV expected CFs + risk adjustment for non-financial risk
- CSM = unearned profit released as insurer provides services
- Assumptions each reporting date = current estimates
- Simplifications/modifications to GMM:
  - PAA for short-duration coverages
  - VFA for investment contracts with discretionary participation features
  - Modified GMM for reinsurance contracts held

**Effective date:** January 1, 2023

- Early adoption concurrent with IFRS 9 (financial instruments) allowed

IFRS 17 presentation requirements?

# Source: ILA101-111: Insurance Contracts First Impressions

- Separate P&L into:
  1. Insurance service result (earnings from insurance components)
  2. Insurance finance income/expense (earnings from investment components)
     - Can be further disaggregated between P&L and OCI
- Insurance revenue reflects changes in liability for future services
  - Excludes investment components and premium

How will IFRS 17 impact how insurers’ earnings are compared and measured?

# Source: ILA101-111: Insurance Contracts First Impressions

- **$\uparrow$ global comparability/transparency of insurers' financial health**
- **Greater volatility in financial results and equity**
  - Key driver: use of current market discount rates
  - $\uparrow$ visibility of economic mismatches between assets/liabilities
  - Product design and/or investment allocation may change in response
- **Key financial metrics will change**
  - Premium volumes no longer key earnings driver
  - Profit emergence may change significantly
- **Clearer earnings picture: separation of financial vs. insurance drivers**
- **Life vs. non-life sector impacts**
  - Bigger impact for life insurers:
    - Significant change: No more locked-in assumptions
    - More visibility on minimum interest guarantees
  - Non-life insurers will want to qualify for PAA to maintain familiarity

List ways IFRS 17 will impact insurers’ processes.

Source: ILA101-111: Insurance Contracts First Impressions

- **New routines**
  - Identifying and accounting for onerous contracts
  - Presenting explicit margin for non-financial risk
  - Accounting for ceded reinsurance separately from direct contracts
- **Communication challenges**
  - New presentation and disclosure requirements
  - New KPIs
  - Education for internal and external users
- **New data, systems, process, and control demands**
  - Key challenges: long time-horizon typical for insurance — legacy systems
  - Will require new controls and upgrades
- **Scarce resources under pressure**
  - IFRS 17 requires significant human talent/resources
- **Opportunities for streamlining and greater efficiency**
- **2nd order impacts: IFRS 17 will trigger new tax/regulatory requirements**

Definition of insurance contract under IFRS 17?

# Source: ILA101-111: Insurance Contracts First Impressions

**Insurance contract** – when one party (issuer) accepts significant insurance risk from another (PH)

- Entitles PH to compensation from issuer when insured event occurs
- Must be enforceable by law
- Doesn't have to be written (can be oral, etc.)
- Doesn't include financial instruments or service agreements
- Includes contracts issued by mutual entities even though PHs bear pooled risk

Define ”insurance risk” under IFRS 17 and provide examples of insurance vs. non-insurance risks.

# Source: ILA101-111: Insurance Contracts First Impressions

**Insurance contracts expose issuer to insurance risk or insurance and financial risk**

**Insurance risk** – non-financial risk transferred from PH to issuer

**Examples of insurance risks:**

- Death or survival
- Injury, illness, or disability
- Property loss due to damage/theft
- Failure to make debt payment when due
- Any other non-financial variable specific to contract party

**Examples of financial risks:**

- Interest rates
- Prices of financial instruments or commodities
- Currency exchange rates
- Price/rate indices
- Credit ratings/indices
- Any other financial variable specific to contract party

Treatment of catastrophe-type non-financial variables under IFRS 17?

**Non-financial variables must be specific to a party to be insurance risks**

- Insurance risks:
  - Contract covering weather event damaging PH's assets
  - Insurance swaps triggered by climate variable specific to contract party
- Not insurance risks (not specific to contract party):
  - Contract based on regional weather index
  - Cat bonds with principal/interest impacted by climate variables

Loan or bond = insurance contract if principal/interest can be adversely impacted by uncertain non-financial events affecting debtor

- Example: Loan balance forgiven on death of debtor
- Insurance contract issuer = bondholder (aka investor or lender)

Treatment of residual value guarantee-type non-financial variables under IFRS 17?

# Source: ILA101-111: Insurance Contracts First Impressions

Insurance contracts if cover risk of changes in FV of specific non-financial asset held by contract party

- Not insurance contracts if only covers risk of changes in market prices

Residual value guarantee on vehicle:

- Insurance contract if:
    - Guarantees residual value of vehicle
    - Amount payable varies with vehicle condition when sold
- Not insurance contract if vehicle must be restored to specified condition

How to assess significant insurance risk under IFRS 17?

# Source: ILA101-111: Insurance Contracts First Impressions

**Insurance risk is significant if a scenario with commercial substance exists where on PV basis, issuer could:**

1. Suffer loss caused by insured event
2. Pay significantly more than if event didn't occur

Additional considerations:

- Assess at contract level
- Insurance risk can be significant even if:
  - Insured event extremely unlikely
  - Expected PV contingent CFs / Expected PV all CFs is small
  - Always consider TVM
  - Significant insurance risk may not be present if issuer can delay timely reimbursement

Additional considerations when assessing insurance risk (review back).

# Source: ILA101-111: Insurance Contracts First Impressions

- **Insurance contract must involve uncertainty with one of:**
  - Probability of insured event
  - Timing of insured event
  - Amount payable if insured event occurs
- **PH must suffer adverse effect to receive compensation**
  - Lapse/persistency and expense risks <u>not</u> adverse to PH
  - Insurer could transfer these risks to another entity to create insurance contract
    - Second entity would apply IFRS 17, but not first
- **Investment contract may become insurance contract in future**
  - Example: deferred annuity allowing annuitization into life annuity
- **Reinsurance contract = insurance contract if either:**
  - Exposes reinsurer to significant loss
  - Transfers substantially all insurance risk on reinsured business

Define investment contracts with discretionary participation features.

# Source: ILA101-111: Insurance Contracts First Impressions

**Investment contract with DPF** pays amounts that are:

1. Expected to be significant portion of total contractual benefits
2. Paid at issuer's discretion (timing or amount)
3. Contractually based on:
   - Returns on specified pool of contracts
   - Realized/unrealized returns on specified pool of assets held by issuer
   - Profit or loss of issuer

NOT insurance contracts, but follow IFRS 17 if issued by entity that
<u>also issues insurance</u>

- If issued by any other entity, follow IAS 32, IFRS 7, and/or IFRS 9

Treatment of fixed-fee service contracts under IFRS 17?

# Source: ILA101-111: Insurance Contracts First Impressions

**Fixed-fee service contract** – contract where service level depends on uncertain event

- These <u>are</u> insurance contracts
- Example: issuer charges fixed fee; agrees to repair car if breaks down

Can apply IFRS 15 if meets these conditions:

- Contract price doesn't reflect individual customer risk
- Compensation = service, not cash
- Insurance risk arises primarily from uncertainty in service usage <u>frequency</u>

IFRS 15 choice made for individual contracts and can't be changed

Treatment of financial guarantee contracts under IFRS 17?

# Source: ILA101-111: Insurance Contracts First Impressions

Reimburse PH for losses when specified debtor fails to make payment

- Usually meets insurance contract definition
- Not insurance if PH doesn't incur loss

IFRS 17 can be optionally applied if issuer regards FGCs as insurance contracts and accounts for them as such

- Election made contract-by-contract and can't be changed

Initial recognition of insurance contracts under IFRS 17 GMM?

Source: ILA101-111: Insurance Contracts First Impressions

**Initial Liability = Initial FCF + Initial CSM**

- FCF = risk-adjusted PV of insurer's rights and obligations to PHs
  - Reflects expected CFs, TVM, and risk adjustment
    - FCF = PV expected CFs + risk adjustment for non-financial risk
    - Expected CFs based on current estimates
  - Risk adj = compensation required to bear uncertainty in amount/timing of non-financial CFs
- CSM = unearned profit recognized as insurer provides services
  - Includes effects of CFs occurring on recognition date
  - Includes de-recognition of any A/L previously recognized before IFRS 17

For profitable groups, Initial FCF $< 0$

- Set CSM = $-$FCF $\rightarrow$ no gain at inception (all profit deferred)

For onerous (non-profitable) groups, Initial FCF $> 0$

- Set CSM = 0 (no profit to defer; recognize loss immediately)
- Amount of FCF called **Loss Component**

Subsequent recognition of insurance contracts under IFRS 17 GMM?

# Source: ILA101-111: Insurance Contracts First Impressions

Liability after inception = sum of:

- Liability for remaining coverage = FCF for future services + remaining CSM
  - Always measured at current estimates
- Liability for incurred claims

Changes in FCF broken into:

- TVM effects: recognize in statement of financial performance
- Changes related to past/current service: recognize in P&L
- Changes related to discount rates: recognize in OCI
- Changes related to future services (e.g. assumption revisions)
  - CSM adjusted to offset these

Interest accretes on CSM each period using original DR

Basic IFRS 17 aggregation requirements?

All contracts must be aggregated into groups on initial recognition:

1. Onerous contracts
2. Contracts with no significant possibility of becoming onerous
3. All remaining contracts

Contracts within each group issued within 1 year of each other

Broad steps in IFRS 17 aggregation process?

# Source: ILA101-111: Insurance Contracts First Impressions

1. **Identify portfolios of insurance contracts** (term, WL, SPDAs, etc.)
   - Portfolio = contracts with similar risks managed together
2. **Group onerous contracts within each portfolio**
   - Can assess in sets only if supportable; otherwise assess individual contracts
   - Contracts priced to be very competitive more likely onerous
   - Identification more challenging if more variation in:
     - Pricing
     - UW practices
     - Contract features/benefits
     - Distribution channel and marketing method
3. **Group profitable contracts based on possibility of becoming onerous**
   - As with onerous, set-level assessment allowed if supportable
   - Possibility of becoming onerous based on:
     - Estimates from entity's internal reporting
     - Likelihood that assumption changes could make contract onerous
   - Analysis requires judgment; focus on most sensitive assumptions
     - Gathering info beyond internal reporting <u>not</u> required
     - Smaller initial CSMs more likely to become onerous

What is the premium allocation approach under IFRS 17?

PAA similar to US GAAP short-duration accounting

- Liability for remaining coverage = unearned premium reduced for acquisition costs
- Assumed to produce results similar to GMM

**Automatically eligible: contracts with coverage periods $\leqslant$ 1 year**

**Other contracts: eligible if PAA measurement doesn't differ materially from GMM**

- Key requirement: FCF can't have significant variability before incurred claims

Many GI/P&C contracts qualify (short coverage periods, no EDs)

Most life insurance contracts do not qualify for PAA

- Insurers may choose to apply GMM to all contracts for consistency anyway

How to determine FCF variability level for PAA eligibility?

# Source: ILA101-111: Insurance Contracts First Impressions

Factors that $\uparrow$ variability in FCFs:

- Existence of embedded derivatives
- Length of coverage period

Considerations when determining FCF variability:

- Key market/risk factors (e.g. interest rates, adjustable premiums)
- Pricing information
- Sensitivity analysis: could establish ”margin of difference”
- Perform analysis more often in unstable interest rate environments

Definition of direct participating contract under IFRS 17?

## Source: ILA101-111: Insurance Contracts First Impressions

**<u>Direct</u> participating contracts** = specific type of participation contract under IFRS 17

- Obligation to PH = FV of Underlying Items − Variable Fee

Insurance contract is **direct participating contract** when:

- **Contractual terms** specify PH participates in share of **clearly identified pool** of underlying items
- Entity expects to pay PH amount = **substantial share** of FV returns on underlying items
- Entity expects **substantial proportion** of change in amounts payable to vary with change in FV of underlying items

Assess criteria at inception and never again unless contract modified

If all criteria met, contract can follow **variable fee approach (VFA)**

Key differences between IFRS 17 GMM and VFA?

# Source: ILA101-111: Insurance Contracts First Impressions

- Changes in FCF arising from TVM and financial risks
  - GMM: Recognize immediately in insurance finance income/expense
  - VFA: Recognize in CSM (provided CSM $> 0$)
    - Insurer's share regarded as part of variable fee
- Interest rate accreted to CSM
  - GMM: Interest rate at initial recognition
  - VFA: None since CSM remeasured for changes in financial risks

Meaning of ”contractual terms” in definition of insurance contract with direct participation features?

# Source: ILA101-111: Insurance Contracts First Impressions

**Link to underlying items must be enforceable by law**

- Sometimes by regulator on behalf of PH
- Agreement doesn’t need to be written: can be oral, electronic, etc.
- Practice can vary across legal jurisdiction and industry

Meaning of ”clearly identified pool of underlying items” in definition of insurance contract with direct participation features?

# Source: ILA101-111: Insurance Contracts First Impressions

- ”Pool” can be anything as long as clearly identified in contract
  - Examples: portfolio of assets or net assets of entity
- Entity not required to hold pool
- Underlying items don’t have to exist for contract duration
- Clearly identified pool does NOT exist when:
  - Entity can retrospectively change underlying items
  - No underlying items identified
- Suppose insurer issues 2 types of contracts with discretionary credited rates:
  - Contract X: Credited rate = 3% currently
    - Does <u>not</u> create link with any clearly identified underlying items
  - Contract Y: Credited rate = return on defined portfolio minus 0.5% margin
    - Does create link

Meaning of ”substantial” in definition of insurance contract with direct participation features?

# Source: ILA101-111: Insurance Contracts First Impressions

**Meaning of ”substantial” in ”substantial share” and ”substantial proportion”**

- ”Substantial” based on how much discretion entity expects to exercise
- Use probability-weighted average across scenarios
  - In some scenarios, FV returns will exceed guaranteed minimums

Treatment of reinsurance under IFRS 17?

# Source: ILA101-111: Insurance Contracts First Impressions

IFRS 17 defines two broad types of reinsurance contracts

1. Reinsurance contracts issued (reinsurance assumed)
   - Follow GMM
2. Reinsurance contracts held (ceded reinsurance)
   - Apply GMM with modifications
   - Account for reins held separately from underlying contracts
   - Obligations for underlying contracts remain after reins

What is the modified GMM for reinsurance contracts held?

# Source: ILA101-111: Insurance Contracts First Impressions

Modified GMM for reinsurance contracts held reflects that:

- Ceded reins usually asset, not liability
- Reins contracts separate from underlying contracts
- Ceding company pays reinsurer premium; reinsurer reimburses ceding company
- Ceding companies usually don’t profit from reins contracts held
- CSM for reins contracts held can be positive or negative
  - Reflects net gain or cost at inception
  - Negative CSM reflects ceded deferred profit

Other considerations for reinsurance contracts held (review back).

## Source: ILA101-111: Insurance Contracts First Impressions

- Reins contracts can never be direct participating contracts
  - I.e. VFA never allowed for any reins contract (held or issued)
- Reins can follow PAA if eligible
  - Assessment separate from underlying contracts
  - Reins coverage may be $>$ 1 year even if underlying contracts $<$ 1 year
- Reins contracts follow same aggregation requirements as underlying contracts
  - May result in group with only one reins contract

Differences in earliest recognition date for proportionate vs. non-proportionate reinsurance under IFRS 17?

**Proportionate reinsurance:** begin recognizing at same date as underlying contracts

- I.e. at least one underlying contract must be recognized first

**Non-proportionate reinsurance:** begin recognizing at earlier of:

- Beginning of reins group’s coverage period
- Date entity recognizes onerous underlying group

TVM accumulation function and properties?

## Source: ICRMELI Ch. 1: Modeling of ELI

Value of money changes over time

$a(t)$ = accumulation function

$$a(0) = 1$$
$$a(t+s) = a(t)a(s) \qquad t, s \geqslant 0$$

Above equality must be satisfied at all times

- If $a(t+s) < a(t)a(s)$, investor better off withdrawing at $t$ and immediately reinvesting
- If $a(t+s) > a(t)a(s)$, investor better off waiting until $t+s$ maturity

How does Law of Large Numbers apply to traditional insurance risks?

LLN: average of many i.i.d. random variables approaches expected value

$$\overline{X}_n := \frac{1}{n}(X_1 + \cdots + X_n) \to \mu, \quad n \to \infty.$$

$X_1, X_2$, etc. can be insurance claims (life, auto, fire, etc.)

Mortality and longevity are **diversifiable risks**

- At policy level, outcomes uncertain
- Uncertainty $\downarrow$ as number of homogeneous policies inforce $\uparrow$

**Key point:** variability of average claim size $\downarrow$ as insurer sells more policies

What is the Central Limit Theorem?

**Lindeberg-Lévy CLT:** average of large number of i.i.d. random variables is approximately normally distributed

$$Z_n = \frac{\bar{X}_n - \mathrm{E}(\bar{X}_n)}{\sqrt{V(\bar{X}_n)}} \to Z, \quad n \to \infty$$

$Z$ = standard normal random variable

What is the Equivalence Premium Principle?

Under equivalence premium principle, $P$ = PV expected benefits only

$$P = \text{PVFB} = \begin{cases} Be^{-rT}\,{}_Tp_x & \text{pure endowment} \\ B \cdot \bar{A}_x = B \cdot \mathrm{E}(e^{-rT_x}) & \text{whole life} \\ Ce^{-rn}\,{}_np_x\bar{a}_{x+n} & \text{deferred immediate annuity} \end{cases}$$

$B$ and $C$ = benefits and annuity payments, respectively

This net single premium $\rightarrow$ expected loss of 0 for each policy:

$$E(L_i) = \text{PVFB} - P = 0$$

What is the Portfolio Percentile Premium Principle?

## Source: ICRMELI Ch. 1: Modeling of ELI

More commonly used than EPP: essentially EPP + margin

$$P = \mathrm{E}(X) + \frac{\sqrt{V(X)}\Phi^{-1}(\alpha)}{\sqrt{n}}$$
$$= \text{Net Single Premium} + \text{Margin}$$

$\Phi^{-1}(\alpha)$ = inverse standard normal CDF at $\alpha$th percentile

- Margin = SD of losses $\times$ number of SDs desired
- $P$ will be sufficient to cover claims $\alpha$% of time
- By CLT, margin $\rightarrow$ 0 as policy count ($n$) $\rightarrow \infty$

Variance of pure endowment (${}_Tq_x = 1 - {}_Tp_x$):

$$\mathrm{V}(X) = Be^{-rT}\,{}_Tp_x\,{}_Tq_x = \mathrm{E}(X) \cdot {}_Tq_x$$

Fundamental principles of Equity-Linked Insurance?

# Source: ICRMELI Ch. 1: Modeling of ELI

- **Traditional Insurance:** LLN diversifies mortality/longevity risk
- **ELI:** Investment risk is **non-diversifiable** (**systematic**)
  - Market dips impact all contracts simultaneously: $\uparrow \max(G - F, 0)$
  - Investment risk remains regardless of $q_x$ certainty
  - Selling more contracts $\uparrow$ exposure to investment risk
- **ELI Violates LLN Assumptions:**
  - **Not Independent:** Contracts correlated via market ($F$ vs $G$)
  - **Not Identical Mean:** Liabilities vary (e.g., by contract size)
    - Wealthy PHs buy larger policies

Sources of investment, revenue, and expenses for deferred VAs?

# Source: ICRMELI Ch. 1: Modeling of ELI

- **Sources of investment** – qualified vs. non-qualified
  - Qualified – tax advantaged funds meeting tax code requirements for retirement plans
  - Non-qualified – after-tax money (e.g. savings account, real estate sale, etc.)
- **Sources of revenue** – fees and charges assessed to PHs
  - GMxB rider charges: % of guarantee base
  - GMDB insurance charges: % of FV
  - Admin expense charges: fixed rate or % of FV, often in M&E fee
  - Investment management fee: % of FV
  - Surrender charge
- **Sources of expenses**
  - Acquisition costs: agent commissions, marketing costs, UW costs
  - Hedging costs: acquiring and managing GMxB hedging portfolios

VA accumulation phase formula using unit calculations?

## Source: ICRMELI Ch. 1: Modeling of ELI

Subaccount value = number of units $\times$ unit value

$$\text{Initial Units} = \frac{\text{Initial Purchase Payment}}{\text{Accumulation Unit Value}}$$

Additional purchases add units; outflows reduce units

Initial UV = arbitrary value, then updated for income, capital gains, M&E fees

At any time, FV = Units $\times$ Unit Value

VA accumulation phase formula ignoring unit calculations?

Subaccount value evolves from time 0 as:

$$F_{k/n} = F_0 \frac{S_{k/n}}{S_0}\left(1 - \frac{m}{n}\right)^k, \quad k = 1, 2, \ldots, nT \qquad \text{(discrete time)}$$

$$F_t = F_0 \frac{S_t}{S_0} e^{-mt} \qquad \text{(continuous time)}$$

PV of insurer's fee income (”margin offset”):

$$M_{k/n} = \sum_{j=1}^{k} e^{-r(j-1)/n}\left(\frac{m_e}{n}\right) F_{(j-1)/n}$$

where $m_e$ = FV-based GMxB charge rate (margin offset)

Possible definitions of guarantee base ($G$) for VAs?

Source: ICRMELI Ch. 1: Modeling of ELI

- **Reset option** – Ensures $G \geqslant F$ immediately after each renewal date ($T_k$)

$$G_{T_k} = \max[G_{T_{k-1}}, F_{T_k}]$$

- **Roll-up option** – $G$ accrues interest at constant rate $\rho$

$$G_{(k+1)/n} = G_{k/n}\left(1 + \frac{\rho}{n}\right) \quad \text{or} \quad G_{k/n} = G_0\left(1 + \frac{\rho}{n}\right)^k$$

- **Step-up option (aka ratchet)** – reset option that can happen <u>any</u> period
  - Lifetime high step-up: $G$ = largest FV to date

$$G_{(k+1)/n} = \max[G_{k/n}, F_{(k+1)/n}]$$

  - Annual high step-up: <u>if</u> FV $\uparrow$, $G \uparrow$ by same rate

$$G_{(k+1)/n} = \frac{G_{k/n}}{F_{k/n}} \max\{F_{k/n}, F_{(k+1)/n}\} \quad \text{or} \quad G_{k/n} = G_0 \prod_{j=0}^{k-1} \max\left\{1, \frac{F_{(j+1)/n}}{F_{j/n}}\right\}$$

- **Other variations**: combinations (max), continuous time, etc.

Common guaranteed minimum benefits sold with VAs?

# Source: ICRMELI Ch. 1: Modeling of ELI

- **GMMB** – Guarantees MB = $\max[G_T, F_T]$ at maturity $T$
- **GMAB** – Guarantees $F = G$ immediately after each reset date $T_k$
  - If $F > G$, $G$ reset to $F$; else $F$ reset to $G$
  - "Payoff" = $\max[G_{T_k} - F_{T_k}, 0]$ at each reset date $T_k$
  - Another variation only adjusts $G$ if $F > G$
- **GMDB** – Guarantees DB = $\max[G_t, F_t]$
- **GMWB** – Guarantees withdrawal benefit, $w$, as % of $G$
  - Withdrawals paid until guarantee fully withdrawn
  - "Ruin time" = point at which FV = 0
  - Takes $T = F_0/w$ periods to deplete guarantee
- **GLWB** – Lifetime version of GMWB
  - Guaranteed withdrawals continue for life even if FV = 0
  - Guarantee charge assessed on $G$, not $F$ like others

Net liability for common VA guaranteed minimum benefits?

# Source: ICRMELI Ch. 1: Modeling of ELI

$L$ = PV Expected Payments In Excess of FV $-$ PV Expected Margin Offset Income

| | Expected Benefits in Excess of FV |
|---|---|
| GMMB | ${}_{T}p_x \max[G_T - F_T, 0]$ |
| GMAB | ${}_{T_k}p_x \max[G_{T_k} - F_{T_k}, 0]$ for each reset date $T_k$ |
| GMDB | ${}_{t-1}p_x q_{x+t-1} \max[G_t - F_t, 0]$ for $t = 1, 2, \ldots, T$ |
| GMWB | $w_t$ for $t >$ ruin time |
| GLWB | ${}_{t}p_x w_t$ for $t >$ expected lifetime |

Interpretation:

- If $L > 0$, $L$ = assets insurer must have to cover future benefits in excess of fees
- If $L < 0$, $|L|$ = PV expected profits
- Sign can change through time and by scenario

Simplified/practical annual FV formulas (review back).

## Source: ICRMELI Ch. 1: Modeling of ELI

For all GMxBs other than GLWB:

$$F_t = \frac{S_t}{S_{t-1}} F_{t-1}(1 - m) - w_t$$

$$m = \text{FV-based charge including GMxB rider charge}$$

$$w_t = \text{withdrawal or surrender amount}$$

For GLWB <u>only</u>:

$$F_t = \frac{S_t}{S_{t-1}} \left[ F_{t-1}(1 - m) - m_w \times G_{t-1} \right] - w_t$$

$$m_w = \text{guarantee-based fee rate applied to } G$$

$$m = \text{FV-based charge not including } m_w$$

How to modify VA formulas for flexible premiums?

Some VAs allow additional contributions, $c$

$$F_{(k+1)/n} = \frac{S_{(k+1)/n}}{S_{k/n}} F_{k/n} \left(1 - \frac{m}{n}\right) + \frac{c}{n}$$

$G$ also $\uparrow$ with each contribution—e.g. for roll-up:

$$G_{k/n} = G_{k/n} \left(1 + \frac{\rho}{n}\right) + \frac{c}{n}$$

Mechanics of immediate variable annuity?

# Source: ICRMELI Ch. 1: Modeling of ELI

Initial calculations:

$$P_0 = \frac{F_0}{\ddot{a}_{\overline{n}|}} \quad \text{and} \quad \text{Units} = \frac{P_0}{V_0}$$

Subsequent calculations:

$$\begin{aligned} P_k &= \text{Units} \times V_k \quad \text{or} \quad P_{k-1} \times V_k \text{ if ignoring units} \\ V_k &= V_{k-1} \times \frac{I_k}{1+i} \\ I_k &= \frac{\text{Previous NAV} + \text{InvInc \& CapGains} - \text{CapLosses \& Charges } (m)}{\text{Previous NAV}} \end{aligned}$$

NAV = PH's net asset value (FV); $I_k$ = net investment factor (NIF)

Formulas used to model immediate variable annuities?

NIF expressed as:

$$I_k = \frac{S_k}{S_{k-1}}(1-m)$$

Allows rewriting formulas for $P_k$ and $F_k$:

$$P_k = P_0 \frac{S_k}{S_0}\left(\frac{1-m}{1+i}\right)^k$$
$$F_k = \frac{S_k}{S_0}(1-m)^k(F_0 - P_0 \ddot{a}_{\overline{k|}})$$

Key differences with life-contingent variable immediate annuities?

# Source: ICRMELI Ch. 1: Modeling of ELI

Life-contingent version based on life annuity factor with AIR

$$P_0 = \frac{F_0}{\ddot{a}_x}$$

$$F_k = \frac{S_k}{S_0}(1-m)^k(F_0 - P_0\ddot{a}_{\overline{k|}}) \qquad \text{(Same as traditional)}$$

**Unlike traditional immediate annuity, LC payments stop immediately on death**

- If PH dies before expected lifetime $K_x$, FV $> 0$ at death
  - Insurer Profit = FV released on death ($F_k$)
- If PH outlives $K_x$, FV = 0, but insurer pays $P_k$ until death

$$\text{Liability} = \sum_{k=1}^{K_x} P_k I(\ddot{a}_{\overline{k|}} > \ddot{a}_x)$$

- Interaction between investments and mortality = more uncertainty for insurer

Compare Fixed-Indexed Annuities (aka EIAs) with VAs.

# Source: ICRMELI Ch. 1: Modeling of ELI

**Key difference with traditional fixed DAs:** more upside through index participation

**Differences with VAs:**

- **Cash Flows:**
  - FIAs: Assets held in GA; insurer bears equity risk
  - VAs: Assets held in SA; PHs bear equity risk (except GMxBs)
- **Risk and Reward:**
  - FIAs: partial participation in equity returns with caps
  - VAs: full participation without caps
- **Fees:**
  - FIAs: no fees; control costs via participation rates, caps, floors
  - VAs: charge fees for management and guarantees (like mutual funds)
- **Regulation:**
  - FIAs: fixed annuities regulated as insurance products
  - VAs: securities regulated by SEC

Common index crediting methods for fixed indexed annuities?

1. **Point-to-Point** – compares index at beginning and end of term
   - Guaranteed min maturity value = $G = Pe^{gT}$
2. **Cliquet** – compares index value at beginning and end of each year
   - Annual index return may be capped at $c$
   - Guaranteed min annual return = $g$ where $g < c < 1$
3. **High-Water Mark** – based on highest index value on policy anniversary
   - Guaranteed min maturity value = $G = Pe^{gT}$

Each design includes index participation rate, $\alpha$

- Can be applied in ”simple” or ”compound” way
- Results similar for small index returns

Maturity value formulas (review back).

# Source: ICRMELI Ch. 1: Modeling of ELI

**PTP maturity value:**

$$\max\left(P\left(1+\alpha\frac{S_T-S_0}{S_0}\right), Pe^{gT}\right) \quad \text{or} \quad \max\left(P\left(\frac{S_T}{S_0}\right)^{\alpha}, Pe^{gT}\right)$$

**Cliquet maturity value:**

$$P\prod_{k=1}^{T}\max\left(\min\left(1+\alpha\frac{S_k-S_{k-1}}{S_{k-1}}, e^c\right), e^g\right) \quad \text{or} \quad P\prod_{k=1}^{T}\max\left(\min\left(\left(\frac{S_k}{S_{k-1}}\right)^{\alpha}, e^c\right), e^g\right)$$

**HWM maturity value:**

$$\max\left(P\left(1+\alpha\left(\frac{\max[S_k]}{S_0}-1\right)\right), Pe^{gT}\right) \quad \text{or} \quad \max\left(P\left(\frac{\max[S_k]}{S_0}\right)^{\alpha}, Pe^{gT}\right)$$

How do forwards and futures work?

# Source: ICRMELI Ch. 4: Pricing and Valuation

Agreement to buy/sell asset at set price (forward price) on future date

- No cost to enter (simply an agreement)
- Forwards typically private and customized
- Futures like forwards but standardized and traded on exchanges

**Payoff**: Contract value at maturity (can be positive or negative)

- Long position payoff = $S_T - F$
- Short position payoff = $F - S_T$

How do call and put options work?

- **Call option**: Gives buyer option to buy $S$ at strike ($K$) by set date
  - Long call payoff = $(S_T - K)^+$
- **Put option**: Gives buyer option to sell $S$ at strike ($K$) by set date
  - Long put payoff = $(K - S_T)^+$
- **Short side:** Negative payoff if long side exercises
  - Else, short's gain = premium received
- **European options**: Exercisable only at maturity
  - **Path-independent**: Payoff depends only on $S$ at maturity
  - Includes PTP FIAs
- **American options**: Exercisable anytime up to maturity
  - **Path-dependent**: Payoff depends on $S$ before maturity
  - Includes cliquet and HWM FIAs

Define arbitrage and no-arbitrage pricing.

# Source: ICRMELI Ch. 4: Pricing and Valuation

**Arbitrage** = trading strategy that:

1. Begins with no money
2. Zero probability of losing money
3. Positive probability of making money

**No-arbitrage principle** or **law of one price** – two transactions with equivalent CFs should have same price

- In efficient market, arbitrage shouldn't exist

**No-arbitrage forward price:** $F = S_t e^{r(T-t)}$

- If $F > S_t e^{r(T-t)}$, borrow to buy asset; enter short side of forward
- If $F < S_t e^{r(T-t)}$, shortsell asset; enter long side of forward

Binomial tree discrete time pricing model?

# Source: ICRMELI Ch. 4: Pricing and Valuation

## Binomial model assumptions

- Only two assets: risky ($S_t$) and risk-free ($B_t$) earning $r$
- $S_t$ can go up by factor $u$ or down by factor $d$ with real-world probabilities:

$$\Pr[S_{t+1} = uS_t] = p = 1 - q$$
$$\Pr[S_{t+1} = dS_t] = q = 1 - p$$

- $d < e^r < u$ (else arbitrage exists)
- **Self-financing:** no external CFs needed
  - No transaction costs
  - Unlimited borrowing/selling/shortselling/buying

## Two equivalent methods for binomial option pricing:

1. Replication portfolio method
2. Risk-neutral probability method

Single period replication portfolio method?

## Source: ICRMELI Ch. 4: Pricing and Valuation

**Goal:** Construct portfolio ($X_0$) replicating option's payoffs in one period

| **Stock Price** | **Option** | **Replication Portfolio** |
|---|---|---|
| $S_0 \begin{cases} uS_0 \\ dS_0 \end{cases}$ | $V_0 \begin{cases} V_1(u) \\ V_1(d) \end{cases}$ | $X_0 = B_0 e^{-r} + \Delta_0 S_0 \begin{cases} B_0 + \Delta_0 u S_0 \\ B_0 + \Delta_0 d S_0 \end{cases}$ |

where $V_1 = (S_1 - K)_+$ for call or $(K - S_1)_+$ for put

**Key steps to calculate option price:**

1. Set $V_1(u) = B_0 + \Delta_0 u S_0$ and $V_1(d) = B_0 + \Delta_0 d S_0$
2. Solve for $\Delta_0 = \frac{V_1(u) - V_1(d)}{S_0(u-d)}$ = portion of $S$ to hold ("delta")
3. Solve for $B_0 = V_1(u) - \Delta_0 u S_0$ <u>or</u> $B_0 = V_1(d) - \Delta_0 d S_0$
4. No-arbitrage option price = $V_0 = X_0$

Single period risk-neutral probability method?

## Source: ICRMELI Ch. 4: Pricing and Valuation

Suppose we can express $S_0$ as risk-free PV of special expected value:

$$\begin{aligned} S_0 &= e^{-r}\tilde{\mathbb{E}}[S_1] \\ &= e^{-r}[\tilde{p}uS_0 + \tilde{q}dS_0] \\ &= e^{-r}[\tilde{p}uS_0 + (1-\tilde{p})dS_0] \end{aligned}$$

$\tilde{p}$ and $\tilde{q}$ = risk-neutral (RN) probabilities ($\neq$ RW probabilities $p$ and $q$)

$$\tilde{p} = \frac{e^r - d}{u - d} \qquad \text{and} \qquad \tilde{q} = 1 - \tilde{p} = \frac{u - e^r}{u - d}$$

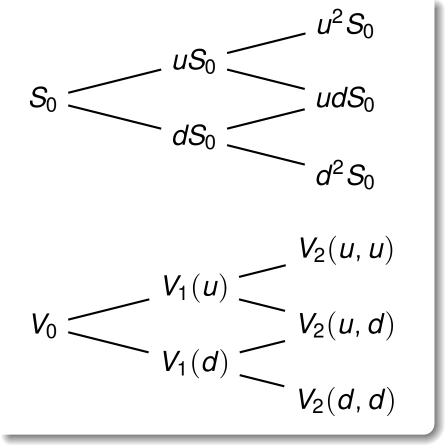
This allows expressing no-arbitrage price as:

$$V_0 = e^{-r}\tilde{\mathbb{E}}[V_1] = e^{-r}[\tilde{p}V_1(u) + \tilde{q}V_1(d)]$$

Under RN framework, all assets expected to yield risk-free rate

$$B_1 = B_0 e^r \qquad \tilde{\mathbb{E}}[S_1] = S_0 e^r \qquad \tilde{\mathbb{E}}[V_1] = V_0 e^r$$

Multi-period risk-neutral probability method?




**Work backwards using single period equations:**

1. At time 1: Solve for 2 possible values of $V_1$

$$V_1(u) = e^{-r}\big[\tilde{p}V_2(u,u) + \tilde{q}V_2(u,d)\big]$$
$$V_1(d) = e^{-r}\big[\tilde{p}V_2(d,u) + \tilde{q}V_2(d,d)\big]$$

2. At time 0: Solve for option price:

$$V_0 = e^{-r}\big[\tilde{p}V_1(u) + \tilde{q}V_1(d)\big]$$

Black-Scholes framework assumptions?

# Source: ICRMELI Ch. 4: Pricing and Valuation

Black-Scholes assumptions similar to binomial model:

- Two assets: risky $S_t$ and risk-free $B_t$ earning $r$
- $S_t$ is continuous variable: follows geometric Brownian motion
- **Self-financing:** no external CFs needed
  - No transaction costs
  - Unlimited borrowing/selling/shortselling/buying

RN formulas for options/derivatives based on "Feymann-Kac formula"

$$f(t,x) = \mathbb{E}[e^{-r(T-t)}h(X_T)|X_t = x]$$

where:

- $f(t,x)$ = value at time $t$ when state variable $X$ has value $x$
- $h(X_T)$ = payoff function at maturity

Black-Scholes RN price for forwards?

# Source: ICRMELI Ch. 4: Pricing and Valuation

No-arbitrage cost to enter forward at inception date $t$:

$$f(t, S_t) = S_t - Fe^{-r(T-t)} = 0$$

Cost of forward at any point after inception:

$$f(t, S_t) = S_t - Fe^{-r(T-t)}$$

Black-Scholes RN price for calls and puts?

## Source: ICRMELI Ch. 4: Pricing and Valuation

No-arbitrage prices of European call and put:

$$c_t = \tilde{\mathbb{E}}[e^{-r(T-t)}(S_T - K)_+ | S_t = s] \qquad p_t = \tilde{\mathbb{E}}[e^{-r(T-t)}(K - S_T)_+ | S_t = s]$$
$$= s\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2) \qquad = Ke^{-r(T-t)}\Phi(-d_2) - s\Phi(-d_1)$$

where:

$$\Phi(z) = \text{standard normal CDF and } \Phi(-z) = 1 - \Phi(z)$$
$$d_1 = \frac{\ln(s/K) + \left(r + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{(T-t)}}$$
$$d_2 = d_1 - \sigma\sqrt{(T-t)}$$

**Put-call parity** relates call and put prices:

$$c_t - p_t = s - Ke^{-r(T-t)}$$

How to calculate distributable earnings on traditional income statement?

## Source: ICRMELI Ch. 4: Pricing and Valuation

1. Calculate Pre-Tax Book Profit = Revenues $-$ Expenses $-$ Benefits
   - Revenues = Premium + InvInc + Policy Fee Income
   - Expenses = Admin Expenses + Commissions + GMDB Cost
   - Benefits = Death Claims + Annuitizations + Surrender Benefits + $\uparrow$ Resv + GMDB Benefit
2. Calculate After-Tax Book Profit = Pre-Tax Book Profit $-$ Tax
   - Tax = Tax Rate $\times$ Pre-Tax Book Profit
3. Calculate $\uparrow$ Target Surplus = EOY Targ Surplus $-$ BOY Targ Surplus
   - EOY Targ Surplus = Targ Surplus Rate $\times$ EOY Resv
   - BOY Targ Surplus = Previous Year EOY Targ Surplus
4. Calculate After-Tax Interest on Targ Surplus = BOY Targ Surplus $\times$Surplus Int Rate $\times$ (1 $-$ Tax Rate)
5. Calculate Distributable Earnings = After-Tax Book Profit $- \uparrow$ Targ Surplus $+$ After-Tax Interest on Targ Surplus

Distributable earnings – simpler income statement (review back).

Pre-Tax Book Profit = Revenues − Expenses − GMDB Benefit

- Revenues = Surrender Charges + Policy Fee Income + Risk Charges
- Expenses = Admin Expenses + Commissions + GMDB Cost

All other steps same as traditional income statement method

Three key profit measures in actuarial pricing?

# Source: ICRMELI Ch. 4: Pricing and Valuation

1. **Net present value (NPV) of distributable earnings**
   - Discount rate = hurdle rate
   - NPV $>$ 0 suggests net profit for insurer; else net loss
2. **IRR = discount rate making NPV = 0**
   - If IRR $>$ hurdle rate, project expected to be profitable
3. **Profit margin (PM)**

$$\text{PM} = \frac{\text{NPV}}{\text{PV of premiums}} \qquad \text{or} \qquad \text{PM} = \frac{\text{NPV}}{\text{PV of Revenues}}$$

**First-year strain:** Large deficit in first year

**Limitation for ELI:** Typically considers only one economic scenario

- Monte Carlo simulations ideal

Compare actuarial vs. no-arbitrage pricing.

# Source: ICRMELI Ch. 4: Pricing and Valuation

- **Actuarial pricing:** based on quantile of profit under RW measure
  - Includes all relevant RW CFs
  - Aims for positive NPV in certain % of RW scenarios: $\Pr[\text{NPV} > 0] > \alpha$
  - Mirrors portfolio percentile premium principle
- **No-arbitrage pricing:** based on expected profit under RN measure
  - Ignores expenses, surrender charges, taxes
  - Expected profit = $\tilde{\mathbb{E}}[L] = 0$
  - Mirrors equivalence premium principle
  - Rarely used in practice

GMAB cash flow profile and impact of fund return?

# Source: ILA101-115: Simulation of a GMAB

- **Policyholder Cash Flows**:
  - Before maturity: Withdrawals/DB per decrement rate
  - At maturity: **gets max(FV, GMAB)**
- **Insurer Cash Flows**:
  - Year 1: Negative from initial expenses
  - Year 2+ before maturity: Positive from M&E and GMAB rider fees
  - At maturity: **pays max(GMAB − FV, 0)**
- **Positive PV** of insurer CFs $\rightarrow$ insurer profitable
- **Impact of Fund Return:**
  - Lower assumed fund return $\rightarrow \uparrow$ likelihood of GMAB payout
  - Very poor (negative) returns $\rightarrow$ large insurer losses
  - GMAB payout can far exceed rider revenue in bad scenarios
  - Risk **not diversifiable** like mortality risk

How do puts hedge a GMAB?

# Source: ILA101-115: Simulation of a GMAB

**Key takeaway:** Hedging significantly stabilizes insurer CFs but at a cost

- Since GMAB = put option, insurer can hedge with put option
- In scenarios triggering GMAB payout, put option also pays off
- Put proceeds help offset GMAB payout
- Put premium = additional insurer expense
- Some volatility remains since fee income sensitive to fund performance

GMAB hedge put option parameters (review back)

# Source: ILA101-115: Simulation of a GMAB

$$
\begin{aligned}
n &= \text{same term as GMAB} \\
K &= \text{CurrentIndex} \times \frac{\text{GMAB}}{\text{Premium} \times (1-m)^n} \\
\text{Notional} &= \frac{\text{Premium} \times (1-w)^n}{\text{CurrentIndex}} \\
\text{Total Put Cost} &= \text{Notional} \times \text{Black-Scholes Put Price}
\end{aligned}
$$

where:

- $m$ = total fee = M&E fee + GMAB rider fee
- $K$ = strike price of single put option on index
  - $m \rightarrow$ higher strike price ($\uparrow$ probability of payoff)
- Notional = number of put options needed
  - Reduced since some PHs lapse ($w$) before GMAB exercised
- Black-Scholes Put Price = price of put on just one unit of index

Considerations for stochastic simulation of GMAB?

# Source: ILA101-115: Simulation of a GMAB

**Problem:** Closed-form solutions usually don't exist (PH behavior, etc.)

**Monte Carlo simulations usually more practical**

- Involves simulating 1000s of RN or RW scenarios
- **Strength of RN:** more accurate for pricing (cost of hedging)
  - Average value more accurate than RW approach
  - Monte Carlo RN average price $\rightarrow$ Black-Scholes price if enough simulations
- **Strength of RW:** provides meaningful distribution of results to analyze
  - Shows asymmetry: most scenarios profitable, but unprofitable scenarios can be <u>very</u> bad
  - Not as accurate as RN since assumed fund return $>$ risk-free rate

Popularity of guaranteed living withdrawal benefits vs. other GLBs?

# Source: VA Guaranteed Living Benefits Utilization

- **GLWB and GMIB most popular by far**
  - 90% of VA GLBs = GLWB or GMIB
  - GLWB has highest sales growth (most premium year 1)
  - GLWB significantly more popular than non-lifetime version (GMWB)
- **GLWB and GMIB have highest persistency (very low surrender rates)**
- GMAB popular with younger buyers

4 inter-connected relationships for GLWB and GMIB utilization?

## Source: VA Guaranteed Living Benefits Utilization

1. **When withdrawals start** (age and source of funding most significant)
   - After age 70, withdrawals $\uparrow$ with age and duration
   - 2/3 of contracts funded with qualified money (tax-favored retirement funds)
2. **Method of withdrawals** (SWPs most significant)
   - SWPs popular for lifetime benefits and older owners (65+)
   - Withdrawals usually continue once started (low surrenders)
3. **Amount of withdrawal taken**
   - Most owners take max allowed by contract (e.g. 4% of benefit base/year)
   - Younger owners more likely to exceed max
   - ITM has little to no effect
4. **Surrender rates – factors $\rightarrow$ low surrenders:**
   - Older owners (65+) in general—even lower if taking withdrawals
   - Younger owners not taking withdrawals yet
   - SWPs have lower surrenders than non-systematic
   - Withdrawal amount $\approx$ 100% of max (not too far below/above)
   - Higher ITM $\rightarrow$ lower surrenders
   - Still in surrender charge period (higher lapse when exiting)

Action steps and issues for pricing VA GLBs?

# Source: VA Guaranteed Living Benefits Utilization

1. **Expect high utilization of lifetime benefits** (hence low lapses)
   - Many used to fund retirement
   - GLB riders popular
2. **Qualified money has highest withdrawal risk**
3. **Withdrawal activity can predict surrender activity**
4. **Modeling GLB utilization important, especially for lifetime options**

Product design comparison of popular VA GLBs?

# Source: VA Guaranteed Living Benefits Utilization

1. **GLWB**
    - Lifetime benefits available ages 50–99 (usually 50s–80s)
    - Median max withdrawal allowed = 4%
    - Benefit base usually reduced proportionally for excess withdrawals
2. **GMIB**
    - Most common design: roll-up or max(roll-up, ratchet)
    - Most are B-share (have SCs)
    - Annuitization rates extremely low ($< 1\%$), but $\uparrow$ with age, size, and ITM
3. **GMAB**
    - Most common benefit base design: 100% of premiums paid in
    - Typical waiting period: 10+ years
4. **GMWB**
    - Differences with GLWB
        - No automatic $\uparrow$ to benefit base if delay withdrawals
        - No cap on benefit base